

DRAFT PROVISIONAL PATENT APPLICATIONS

Series A: the twenty-six JoinAQAL sister combinations. Every combination in this series is implemented and operating in the JoinAQAL codebase — software reduction to practice.

STATUS: DRAFT — NOT FILED. No application numbers exist; nothing herein is patent-pending. Per company records; confirm status with patent counsel. Prepared with AI assistance; not legal advice; attorney review required before filing.

Inventor: Samuel Andrew Russell V — Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-01

Title of Invention: Consensus-Driven Coaching Synthesis

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the consensusDrivenCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-02

Title of Invention: Transparent Integral Framework

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* `server/patents/combinations/index.ts` (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* `server/patents/ledger.ts`; `/api/ledger/verify`; `/api/ledger/head`.

Combinator. The combination executes as the `transparentIntegral` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
5. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
6. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-03

Title of Invention: Voice-Based Weakness Identification

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Combinator. The combination executes as the voiceWeakness combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production

scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-04

Title of Invention: Voice-Transparent Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* `server/patents/ledger.ts`; `/api/ledger/verify`; `/api/ledger/head`.

Combinator. The combination executes as the `voiceTransparentConsensus` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or

reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-05

Title of Invention: Staged Weakness Identification Engine

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Combinator. The combination executes as the stagedWeakness combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus

(per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-06

Title of Invention: Coaching Transparency Framework

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the coachingTransparency combinator (server/patents/combinations/index.ts), registered in the sister-combination registry

(server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

5. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

6. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-07

Title of Invention: Voice-Driven Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping

carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising:
(a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted

timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-08

Title of Invention: Transparent Staged Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Developmental Stage-Band Estimator, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash

commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentStagedConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-09

Title of Invention: Integral Voice Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Integral Four-Quadrant Mapper. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Combinator. The combination executes as the integralVoiceConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.
6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-10

Title of Invention: Staged Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the stagedWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising:
(a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-11

Title of Invention: Transparent Integral Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify;

/api/ledger/head.

Combinator. The combination executes as the transparentIntegralCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or

treatment claims.

4. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

5. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

6. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-12

Title of Invention: Voice-Driven Staged Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceStagedWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average;

(d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-13

Title of Invention: Transparent Integral Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit,

deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentIntegralConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted

timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-14

Title of Invention: Staged Integral Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the stagedIntegralWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-15

Title of Invention: Consensus-Driven Integral Coaching with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Integral Four-Quadrant Mapper, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping

carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the consensusIntegralCoachingTransparency combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the

recommendation chain is traceable from score to protocol to source; (d) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-16

Title of Invention: Voice-Driven Staged Integral Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* `server/patents/combinations/index.ts` (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* `server/scoring/controllingWeakness.ts`.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* `shared/therapyLineMap.ts`; `server/patents/weaknessInterventions.ts`.

Combinator. The combination executes as the `voiceStagedIntegralWeaknessCoaching` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric,

so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (d) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (e) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (f) combining the outputs of steps (a)–(e) into a single emergent result not produced by any component engine alone; and (g) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-17

Title of Invention: Full-Spectrum Integral AI Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine, Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per

assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the fullSpectrumIntegralCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the

member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (f) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (g) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (h) combining the outputs of steps (a)–(g) into a single emergent result not produced by any component engine alone; and (i) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.
6. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
7. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a

confidence threshold, and floors only rise.

8. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-18

Title of Invention: Adaptive Voice Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Combinator. The combination executes as the adaptiveVoiceConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.
6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-19

Title of Invention: Transparent Voice Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentVoiceCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component

engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-20

Title of Invention: Integral Weakness Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:*

server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Combinator. The combination executes as the integralWeaknessConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-27

Title of Invention: Voice-Driven Cognitive Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceDrivenCognitiveCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-28

Title of Invention: Transparent Cognitive Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentCognitiveAssessment combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising:
(a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-30

Title of Invention: Voice-Driven Cognitive Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Combinator. The combination executes as the voiceDrivenCognitiveAssessment combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising:
(a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.
6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
8. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-34

Title of Invention: Cognitive-Financial Development Tracker

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the `cognitiveFinancialDevelopmentTracker` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the `JoinAQAL` codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising:
(a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-36

Title of Invention: Voice-Driven Cognitive Coaching with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the voiceCognitiveCoachingTransparent combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component

engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-38

Title of Invention: Voice-Driven Cognitive Assessment with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the voiceCognitiveAssessmentTransparent combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

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Claims (DRAFT — for attorney revision)

1. A computer-implemented method for producing an emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of

history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, preventing false pitch detection in silence, and wherein the raw audio never leaves the server for the claimed analysis.

6. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

7. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

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